Towards Synthetic Cryptography
A String Diagrammatic Approach

Martti Karvonen[1,2], Robin Piedeleu[1], and Fabio Zanasi[1]

[1] University College London, London, UK
[2] University of Bath, Bath, UK

Abstract Security proofs in cryptography are organised as sequences of games, each indistinguishable from the next by an adversary. Individual steps are local – a component is replaced, a bad event ruled out, an algebraic identity applied – yet they sit inside large global descriptions of the games involved, which obscures the compositional structure of cryptographic constructions and makes proofs hard to verify, reuse, and mechanise. We report on work in progress that recasts game-based cryptography in the language of string diagrams, giving the circuit-like pictures cryptographers already draw a formal status as objects of reasoning. We give these diagrams a compositional semantics in terms of streams over efficiently computable substochastic maps, a model in which two notions of equivalence coexist – stream equality and computational indistinguishability. This semantics makes diagrammatic security arguments precise: proofs become sequences of local, checkable rewrites, while the cryptographic principles they invoke (*e.g.*, pseudorandomness, the birthday bound, the one-time pad) are isolated as axioms. We demonstrate the framework by reconstructing the security proof of the 3-round Feistel cipher, and set out the research programme it opens.

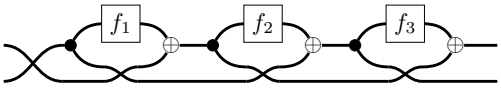
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1 Introduction

Modern digital infrastructure runs on cryptographic guarantees, yet the *proofs* that establish them are becoming harder to produce, audit, and reuse as protocols grow. In the computational paradigm, the dominant setting of modern cryptography, security is defined through a *game* between an adversary and a challenger, and a construction is deemed secure when every efficient adversary has only negligible advantage in the corresponding game. Such guarantees are established by *hybrid arguments*: long sequences of game transformations in which each consecutive pair is shown indistinguishable, combining algebraic, probabilistic, and bookkeeping steps across many rounds and oracle queries. While effective, this style obscures the compositional structure underlying many cryptographic arguments: most proof steps are inherently *local*, yet they sit within large global game descriptions, making proofs difficult to verify, reuse, or mechanise.

As cryptographic systems are layered into increasingly complex stacks (post-quantum migrations, secure messaging, multi-agent systems), this gap between the modular structure of constructions and the global structure of their proofs becomes a bottleneck for scaling trust. Addressing it calls for methods of establishing cryptographic guarantees that scale alongside the systems they concern: a prerequisite for large-scale machine-assisted cryptographic reasoning.

At the same time, cryptographic constructions are themselves already described compositionally, a feature not yet exploited at a formal level. A Feistel network, for instance, is obtained by wiring together copies of a round pseudorandom function with XOR gates and swaps. Cryptographers routinely depict such constructions using circuit-like diagrams that make the flow of information explicit. For the 3-round Feistel cipher, the picture



conveys the dataflow at a glance. Such diagrams give valuable intuition about composition, but currently play only an informal, explanatory role.

Aims. We take this notation seriously as a formal language: the circuit-style diagrams already used by cryptographers become a language of *string diagrams*, equipped with a compositional semantics capturing their intended computational behaviour. Security proofs then become equational derivations in this language: sequences of local diagrammatic transformations justified by algebraic identities, probabilistic principles, or cryptographic assumptions formulated as axioms. This yields a *synthetic* account of game-based cryptography, echoing Fritz's synthetic approach to probability [15]: security arguments are carried out axiomatically, within the diagrammatic language, instead of by analytic computation in a particular semantic model. Composition is a primitive, so replacing a component is a local rewrite with no ad hoc changes to a global game; the syntax is *resource-sensitive*, making the copying and discarding of data – delicate in the presence of randomness – explicit; and because syntax and semantics are cleanly *separated*, one syntax can be interpreted in different computational models, pointing towards a substrate-independent foundation for trust.

Contributions. This work-in-progress articulates a categorical perspective on game-based cryptography and demonstrates its feasibility on a central example, stopping short of full generality. We outline (i) a string diagrammatic syntax for cryptographic constructions (Section 2); (ii) a stream-based categorical semantics distinguishing two notions of equivalence: equality of behaviours (established coinductively) and computational indistinguishability (established inductively); and (iii) a diagrammatic reconstruction of the Luby–Rackoff theorem, in which the security of the 3-round Feistel cipher is proved as a sequence of local rewrites (Section 3). To our knowledge this is the first diagrammatic treatment of a non-trivial *computational game-based* proof, providing evidence that the approach is expressive enough to capture standard cryptographic arguments, and not only in the composably secure setting studied previously. We then lay

out the research programme this opens up (Section 4). The details on the syntax, semantics, and diagrammatic derivations are developed in the companion report [18].

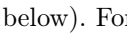
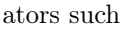
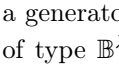
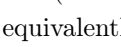
Related work. Mechanised cryptographic verification has produced expressive logics for game-based proofs in the computational model, including EasyCrypt [2], CryptoVerif [5], and Coq- and F*-based frameworks [3, 35], which typically interpret games as programs. Closest in spirit is EasyCrypt's probabilistic relational Hoare logic, reasoning about program equivalence and indistinguishability through sequences of program transformations. Symbolic tools such as ProVerif [6] and Tamarin [4] analyse protocols in the Dolev–Yao model [13], at a different level of abstraction; composable frameworks such as universal composability [11] and abstract cryptography [25] bring algebraic structure to reductions and composition.

Our work is complementary to these approaches. Rather than introducing a new program logic or verification environment, we aim to identify the algebraic structure underlying game-based security arguments. The goal is a more synthetic account of these proofs, in which the compositional principles guiding hybrid arguments become explicit and amenable to algebraic reasoning. In doing so, we were inspired by the informal but insightful blog post by Meier [26]. Existing pictorial representations in cryptography [31] remain informal illustrations; earlier string-diagram work in cryptography [7, 28] was confined to quantum, symbolic, or fixed-size settings, leaving asymptotic notions such as pseudorandomness out of reach. Closest to our setting, the categorical composable cryptography of [8, 9], by one of the present authors, addresses composable security; here we target the setting of *game-based* proofs.

2 Methodology: A Diagrammatic Calculus

String diagrams. To make circuit-based reasoning formal we adopt *string diagrams*, a graphical language from category theory that generalises the circuit diagrams already familiar to cryptographers and supports formal reasoning across program semantics, machine learning, quantum theory, and circuit languages (see [30] for a recent survey). Components are drawn as boxes connected by wires, with sequential and parallel composition expressed spatially rather than symbolically. String diagrams are the syntax for monoidal categories; they can then be equipped with a semantics by giving a monoidal functor into some monoidal category capturing the desired class of behaviours. Indeed, the underlying structure is *monoidal* rather than *cartesian*: because cryptographic constructions rely crucially on randomness, copying and discarding data are explicit operations that may be forbidden, where ordinary syntax would duplicate and delete variables for free. The *resource-sensitivity* coming from the presence of randomness makes string diagrams well suited to probabilistic computation, quantum theory, and cryptography alike; reformulating the same arguments in classical syntax would obscure it and force one to track linear versus non-linear substitution and α-equivalence by hand. Structural facts about a construction thereby become

geometric facts about its diagram. The trajectory of the ZX-calculus is a direct precedent: introduced for reasoning about quantum processes, it now underpins standard tools for quantum-circuit compilation and optimisation [1, 14, 19, 21].

Syntax. Our approach to identifying syntactic cryptographic primitives draws inspiration from previous categorical accounts of cryptography [7–9], as well as diagrammatic calculi for probabilistic computation [15, 29], and programming-oriented presentations of game-based cryptography [31]. Cryptographic constructions are built as string diagrams from a collection of generators. Wires carry values of a given type: finite sets such as bitstrings $\mathbb{B}^\lambda$ of a length fixed by the security parameter λ, and tuples thereof. We further distinguish *thin* wires, that may be used once, from *thick* wires that can be used at every step of the interaction with the adversary, in effect representing a *stream* of wires (see *Semantics* below). For example,  represents the identity over $\mathbb{B}^\lambda$ at every step of the computation. Beyond the boxes representing primitives (Boolean operations, a uniform distribution generator ), the calculus includes structural generators such as copying  , discarding  , and wire-crossings. We also have a generator  , representing a stateful operation which takes in an input of type $\mathbb{B}^\lambda$ at every step and returns the same input if it has not seen it before (and fails otherwise), and  , which denotes a random dictionary or, equivalently, a random function of type $\mathbb{B}^\lambda \to \mathbb{B}^\lambda$. Diagrams can be composed sequentially $(;)$ if their types match and in parallel $(\otimes)$.

Semantics. To model cryptography in the computational model, diagrams with thin wires are first interpreted in the symmetric monoidal category of *efficiently computable substochastic maps*: sequences of subdistribution-valued maps indexed by a security parameter λ, implementable by a probabilistic Turing machine running in time polynomial in λ [9]. Cryptographic protocols, however, unfold over multiple rounds and maintain state, and adversaries query components adaptively. To interpret diagrams with thick wires, we therefore move to *stateful* maps that maintain a memory, and then to *streams*, which forget the internal state and record only the observable input/output behaviour over time. Our treatment builds on recent categorical accounts of stateful and stream-based computation [12], adapted to the setting of modern cryptography. *Equality of streams* $(=)$ is defined *coinductively*: streams are specified by their head and tail, and equalities are derived by unfolding them one step at a time, through stream-unfolding axioms (see Section 3) [17, 33]. Several components of our example, such as the random dictionary, are themselves defined coinductively in this way.

However, equality of streams is insufficient to capture the semantics of many cryptographic constructions, which are defined with respect to a resource-bounded adversary. For this reason, we introduce a second equivalence relation on streams, *computational indistinguishability* $(\approx)$, defined relative to polynomial-time adversaries making polynomially many adaptive queries [18, Definition 8]. We show that computational indistinguishability is a congruence for the sequential and

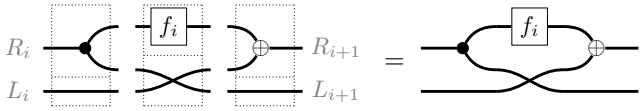
parallel composition of streams, strictly coarser than stream equality [18, Theorem 1].

This licenses rewriting inside an arbitrary context for both forms of equivalence: reasoning proceeds as a two-dimensional generalisation of ordinary algebra, where to apply an identity $l = r$ (or $l \approx r$) one locates l as a subdiagram and replaces it by r.

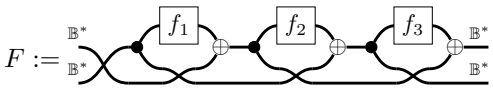
3 Case Study: the 3-Round Feistel Cipher

To demonstrate the approach we analyse a central case study: the security of the 3-round Feistel construction, originally proved by Luby and Rackoff [23]. This example is simple enough to describe succinctly, yet its security proof is non-trivial and already exhibits the essential features of modern cryptographic reasoning. Our argument follows the structure of the proof presented by Rosulek [31, Theorem 7.3.4], recast in compositional diagrammatic form.

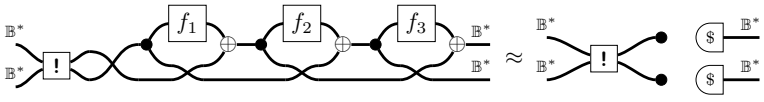
The construction. A pseudorandom function (PRF) is a family $f = \{f_\lambda \colon \mathbb{B}^\lambda \times \mathbb{B}^\lambda \to \mathbb{B}^\lambda\}_{\lambda\in\mathbb{N}}$ that is computationally indistinguishable from a truly random function when given a uniformly random key in $\mathbb{B}^\lambda$; in our notation, $f \circ (\mathsf{id} \otimes \$) \approx R$. The Feistel cipher constructs a pseudorandom *permutation* from a PRF: one round of the cipher maps (L_i, R_i) with key K_i to $L_{i+1} = R_i$ and $R_{i+1} = L_i \oplus f_i(R_i)$, where $f_i(-) := f(-, K_i)$; the 3-round cipher F composes three such rounds. As a string diagram each round decomposes transparently into copy, the round function, a swap, and XOR,



Since the environment may call the cipher several times, all components here represent streams over $\mathbb{B}^\lambda$ and are thus drawn with thick wires. The 3-round cipher is obtained by sequencing three rounds with independent keys:



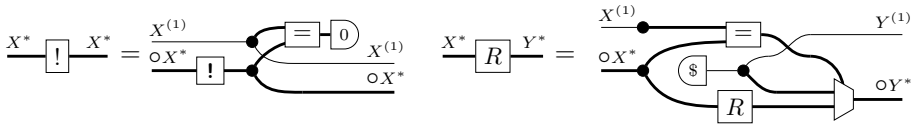
It is easily checked that the diagram above represents a permutation; we must show that it is pseudorandom. Since a random permutation is indistinguishable from a random function queried on distinct inputs (the PRP/PRF switching lemma, see *e.g.* [31]), it suffices to show that F is indistinguishable from a random function on distinct inputs:



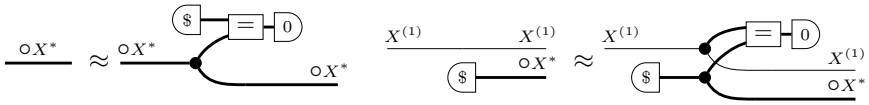
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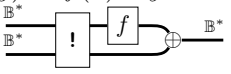
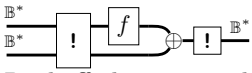
where, in this case, the **!**-box constrains its input to be a stream of distinct *pairs*, *i.e.* distinct elements of $\mathbb{B}^\lambda \times \mathbb{B}^\lambda$. Our goal is to *derive* (1) by equational reasoning with diagrams.

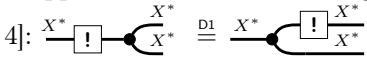
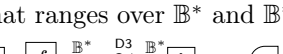
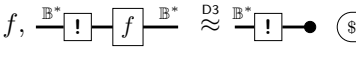
Axioms. We isolate a small collection of identities. First, working with the well-understood theory of substochastic maps as the ground layer lets us reuse existing complete axiomatisations of probabilistic operations [29] (e.g. the one-time pad ). Besides these, we use two *stream-unfolding* axioms, that characterise the behaviour of the **!**-box and the random dictionary R equationally. They are presented in the style of Rutten's stream calculus [33] with the head on the top thin wire labelled by $X^{(1)}$ and the tail on the bottom thick wire labelled by $\circ X^*$ [18, Definition 5]. Unfolding the **!**-box forces the first input to be distinct from all subsequent inputs, themselves subjected to the same constraint. The random dictionary R answers its first query with a freshly sampled value; its tail returns that same value whenever the first query recurs, and otherwise behaves as a fresh random dictionary:



The next two axioms are more clearly cryptographic in nature and involve computational indistinguishability explicitly. They capture the *birthday bound*, which states that a freshly sampled value from a sufficiently large set X (one whose cardinality grows faster than any polynomial in λ) is, with overwhelming probability, distinct from any previously seen value:



We then require a *uniqueness* axiom expressing that $(x, y) \mapsto f(x) \oplus y$ behaves as a universal hash when f is a PRF, *i.e.*, when $f \approx R$:   . This axiom contains the technical heart of the Luby–Rackoff theorem and we refer the reader to Rosulek's argument [31, Theorem 7.3.4] for a proof of its soundness, leaving a fully diagrammatic derivation (from simpler principles, such as stream equivalences and the birthday bound) for future work.

With these axioms in place, our framework reconstructs Rosulek's argument [31] as a sequence of local transformations on diagrams. Figure 1 traces the resulting derivation, with each rewrite labelled by the axiom or derived law it applies. We use the following derived laws [18, Lemma 2, Lemma 3, Lemma 4]:  (X here is arbitrary and D1 will be used below for **!** that ranges over $\mathbb{B}^*$ and $\mathbb{B}^* \times \mathbb{B}^*$),  and, for any PRF f,  .

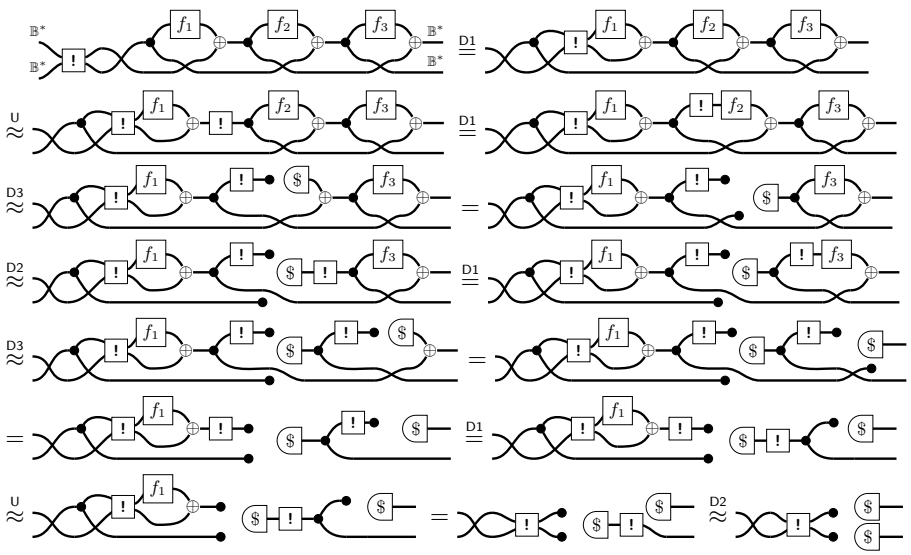


Figure 1. Diagrammatic proof of security of the 3-round Feistel construction. The **!**-box enforces distinct inputs; the derivation establishes that on distinct inputs the cipher's outputs are indistinguishable from uniform (\$-boxes, final diagram). Unlabelled steps are equalities between substochastic maps, derivable in the complete calculi of [16, 29]; the rest are labelled with the axiom or derived law applied.

Theorem 1 (Luby–Rackoff, diagrammatically). *The 3-round Feistel cipher built from pseudorandom round functions using independent keys is a pseudorandom permutation; equivalently, equation* (1) *is derivable from the axioms above.*

The result itself is classical; what our reconstruction changes is the *form* of the proof: a monolithic argument becomes a sequence of local, reusable steps, with the structural, probabilistic, and cryptographic ingredients cleanly separated.

4 Future Work

Axiomatic calculi for cryptographic proofs. A natural next step is to extend the methodology to a broader class of constructions such as arguments involving several PRF invocations using a shared key (*e.g.*, CBC-MAC), symmetric-key cryptography, hashing, authentication [31]. The central technical challenge is to isolate, for each construction, which equalities are *structural* (laws of monoidal categories, dataflow identities), which arise from *probabilistic reasoning*, and which depend on *asymptotic or cryptographic assumptions* (birthday-type bounds, pseudorandomness), making each explicit as an axiom or inference rule. We expect many arguments to reduce to combinations of stream-unfolding laws and a handful of probabilistic approximations, yielding a reusable library of diagrammatic proof principles in which standard steps are packaged as compositional rewrite rules, much as state-separating proofs modularise code-based

game-playing arguments [10], though how far this kind of reduction extends remains to be seen. Such a library would also lend itself to mechanisation in diagrammatic proof assistants such as chyp [20].

Comparing cryptographic models. Beyond individual constructions we would like to isolate which aspects of security arguments are model-dependent. Because our framework separates syntax from semantics, the same diagrammatic language can be interpreted in different symmetric monoidal categories – efficiently computable substochastic maps for the computational model, unrestricted stochastic maps for information-theoretic settings, algebraic terms for symbolic Dolev–Yao models [13], completely positive maps in a quantum setting. This suggests comparing models *functorially*, asking which diagrammatic equalities and indistinguishability relations are preserved when the interpretation changes. Frameworks for composable cryptography [8, 9, 11, 24, 25] share this model-relative perspective, though they target composable security rather than the cross-model transfer of axioms. Relationships between information-theoretic, computational, and quantum security could then be studied axiomatically, and where axioms fail to transfer the analysis may yield general no-go results. The long-term ambition is a substrate-independent diagrammatic core, with model-specific assumptions appearing explicitly as axioms.

Quantitative reasoning about distinguishing advantage. Strict, yes/no indistinguishability is often too rigid: one must quantify an adversary's distinguishing advantage, and many constructions are only approximately secure. We aim to enrich our semantic category with a pseudometric tracking *how far* two streams are from one another. On the syntactic side this leads to *quantitative* equational reasoning with diagrams, as developed in [22, 34], where a judgement $d \approx_\epsilon d'$ guarantees that $[\![d]\!]$ and $[\![d']\!]$ differ by at most ϵ. Even the birthday bound could then be derived from the statistical distance between sampling with and without replacement. Such a calculus would also support tighter security bounds, in the spirit of Patarin's H-coefficient technique [27].

Coinductive characterisations of indistinguishability. A related challenge is the nature of computational indistinguishability itself. Interactive systems are modelled as *streams*, captured categorically as *coalgebras* [32] – here the coinductive streams of [12] – whose equality is established by *coinduction*: exhibiting a bisimulation that relates their step-by-step behaviour [17, 33]. Computational indistinguishability, by contrast, quantifies over polynomial-time distinguishers, so the two equivalences are presently established by different means. The open question is whether indistinguishability admits a coinductive, bisimulation-based characterisation, likely a *quantitative* one parameterised by the number of interaction steps and the advantage accumulated over them; such an account would place both equivalences on a common footing.

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